

Please enter what you learned from today's lecture in at least 30 words.

Today I learned how crucial feature engineering is for preparing data before training a model. We covered transforming numerical variables using scaling and log transformations, and converting categorical data using label and one-hot encoding. We also dived deep into ensemble learning, discovering how bagging (like Random Forests) reduces variance while boosting (like XGBoost) reduces bias, dramatically improving model accuracy.

Please provide specific details about what you particularly liked about today's lecture.

I really loved the practical demonstration using the Titanic dataset. Showing how combining "Sex" and "Pclass" into a single cross-feature revealed that 1st-class women survived at about 97%, while 3rd-class men survived at 14% was incredibly eye-opening. It perfectly illustrated why using domain knowledge to engineer features is sometimes much more powerful than relying on raw data alone.

If there were any parts that were unclear or areas for improvement, please provide specific details.

The concepts of the bias-variance tradeoff and polynomial regression were a little bit fast. While the U-curve makes sense, spending just a bit more time breaking down the math or showing more visual animations of how bagging and boosting specifically affect the U-curve would be super helpful for beginners.

Regarding the instructor, please provide specific details about what you liked or disliked.

Shinazawa-sensei was great! I liked that he shared a bit about his personal background, like studying in New Zealand and playing the cello, which made the lecture feel more welcoming and human. He also did a fantastic job during the Q&A session explaining why logistic regression is sensitive to scaling, while tree-based models are not.

Please write down any lectures or fields you would like to attend in the future.

I would love to learn more about unsupervised learning, which was mentioned briefly as an upcoming topic. I'm also really interested in deep learning and natural language processing.

Please feel free to write any other comments or opinions.

Thanks for the great session! The notebook exercises were incredibly well-structured, and having the interactive hands-on portion right after the video lecture made it much easier to digest all the heavy concepts.